

Ramsey Theory

Caltech Math Meet 2025

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Motivating problem

Example

What is the smallest number of people needed in a room to guarantee that three people are mutual friends or mutual strangers?

Definition of a graph

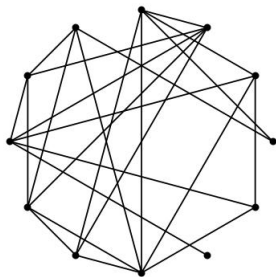
Definition

A *graph* $G = (V, E)$ is a collection of *vertices* and *edges* or pairs of vertices.

Definition of a graph

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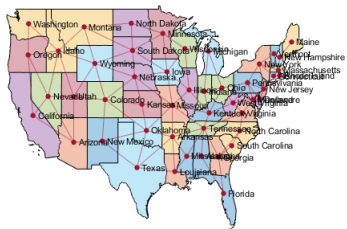
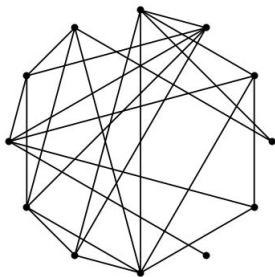
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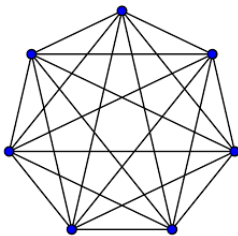
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Complete graph

Definition

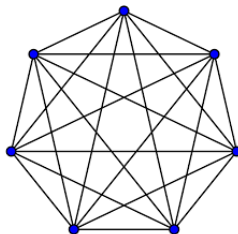
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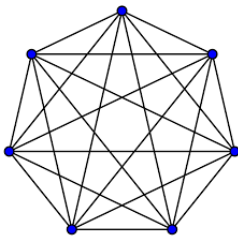


How many edges does the complete graph on n vertices have?

Complete graph

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How many edges does the complete graph on n vertices have?

$$|E(K_n)| = \frac{n(n-1)}{2}$$

Edge-coloring of complete graphs

Example

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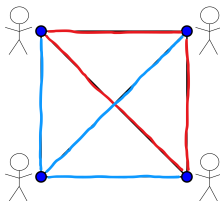
If we represent each person as a vertex in a complete graph, we can color an edge blue to denote friends and red to denote strangers.

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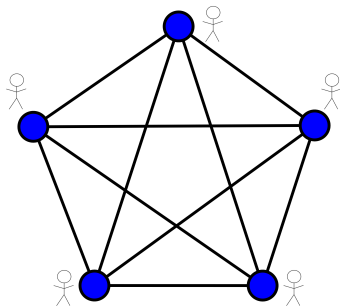


Top left, top right, and bottom right are mutual strangers.

Motivating problem

Example

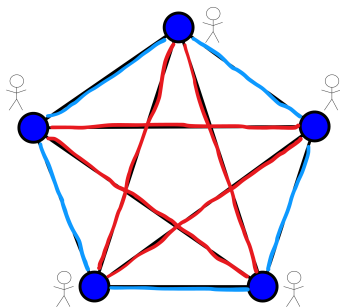
What is the smallest n for which any 2-coloring of the edges of K_n yields a monochromatic triangle?



Motivating problem

Example

What is the smallest n for which any 2-coloring of the edges of K_n yields a monochromatic triangle?



Pigeonhole principle

On the previous slide, we argued that the answer to our motivating problem is more than 5. We now want to prove that any 2-coloring of the edges of K_6 will have either a blue triangle or red triangle.

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Proposition (Pigeonhole principle)

If n items are put into m containers with $n > m$, then at least one container must contain more than one item.

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Motivating problem

Lemma

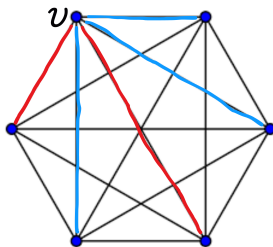
Any red/blue-coloring of the edges of K_6 will either have a blue triangle or a red triangle.

Motivating problem

Lemma

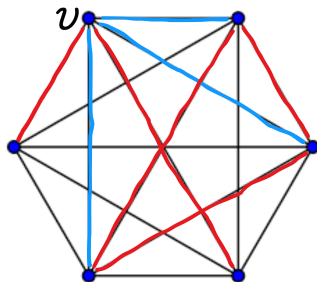
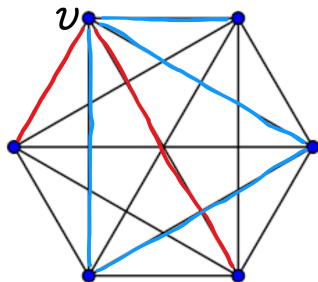
Any red/blue-coloring of the edges of K_6 will either have a blue triangle or a red triangle.

Think of edges out of v as pigeons and colors as pigeonholes. By pigeonhole principle, v has at least three red neighbors or at least three blue neighbors (pictured).



Motivating problem

Vertices connected to v by blue edges form the *blue neighborhood*.



Any blue edge between vertices in the blue neighborhood results in a blue triangle (left picture). If all the edges between vertices in the blue neighborhood are red, we have a red triangle (right picture).

Ramsey numbers

Proposition

The smallest n for which any 2-coloring of the edges of K_n yields a monochromatic triangle is $n = 6$.

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The *Ramsey number* $R(s, t)$ is the smallest n for which a 2-coloring of the edges of K_n must contain a blue K_s or a red K_t .

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The *Ramsey number* $R(s, t)$ is the smallest n for which a 2-coloring of the edges of K_n must contain a blue K_s or a red K_t .

The motivating problem was to find $R(3, 3)$ since a triangle is a complete graph on three vertices.

Compute Ramsey number $R(2, t)$

Example

Compute $R(2, t)$ for any integer $t \geq 2$.

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$$R(2, t) = t$$

Color each edge of K_{t-1} red. The graph will contain neither a blue K_2 nor a red K_t .

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Compute $R(2, t)$ for any integer $t \geq 2$.

$$R(2, t) = t$$

Color each edge of K_{t-1} red. The graph will contain neither a blue K_2 nor a red K_t .

If we color any edge blue in K_t , we are guaranteed a blue complete graph on two vertices. If we color all edges of K_t red, then we have a red copy of K_t .

Ramsey's Theorem and Erdős–Szekeres inequality

Theorem (Ramsey's Theorem)

There exists a least positive integer $R(s, t)$ for which every 2-coloring of the edges of the complete graph on $R(s, t)$ vertices contains a blue K_s or a red K_t .

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For integers $s, t \geq 2$, $R(s + 1, t + 1) \leq R(s, t + 1) + R(s + 1, t)$.

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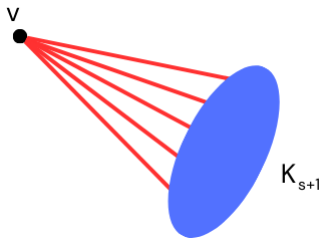
Corollary (Erdős–Szekeres, 1935)

For all natural numbers $s, t \geq 1$, $R(s + 1, t + 1) \leq \binom{s+t}{s}$.

Erdős–Szekeres inequality

Proof.

Color each edge of $K_{R(s,t+1)+R(s+1,t)}$ red or blue. Choose vertex v . If the red neighborhood of v has at least $R(s+1, t)$ vertices, the neighborhood contains a blue K_{s+1} or a red K_t . Adding v to the red K_t produces a red K_{t+1} . Make a similar argument if the blue neighborhood of v has at least $R(s, t+1)$ vertices. $\square$



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Erdős–Szekeres inequality implies $R(4, 3) \leq R(3, 3) + R(4, 2) = 10$, but pigeonhole principle improves the upper bound to $R(4, 3) \leq 9$.

Erdős–Szekeres inequality

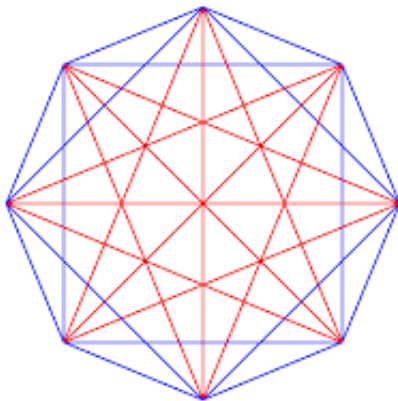
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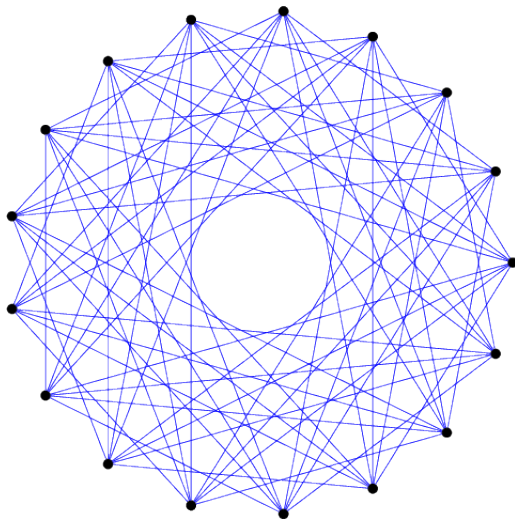
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Erdős–Szekeres implies $R(4, 4) \leq R(3, 4) + R(4, 3) = 18$. The following illustrates blue edges in a 2-coloring of K_{17} that contains neither a blue K_4 nor a red K_4 . We conclude $R(4, 4) = 18$.

Erdős–Szekeres inequality



Why are Ramsey numbers difficult to compute?

“Erdős asks us to imagine an alien force, vastly more powerful than us, landing on Earth and demanding the value of $R(5, 5)$ or they will destroy our planet. In that case, he claims, we should marshal all our computers and all our mathematicians and attempt to find the value.

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But suppose, instead, that they ask for $R(6, 6)$. In that case, he believes, we should attempt to destroy the aliens.”

–Joel Spencer

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There are $\frac{n(n-1)}{2}$ edges. Each edge can be red or blue, so there are $2^{\frac{n(n-1)}{2}}$ ways to color the edges of K_n . Many of these edge-colorings are the same, but even determining repeats is an unsolved problem.

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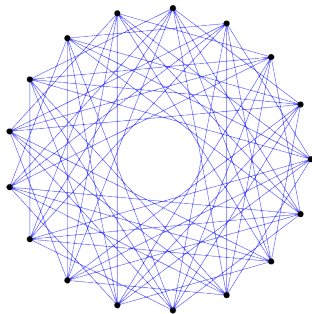
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According to some estimates, there are roughly 10^{80} atoms in the observable universe. The complete graph on 26 vertices already has more red/blue-colorings of its edges than there are particles in the observable universe.

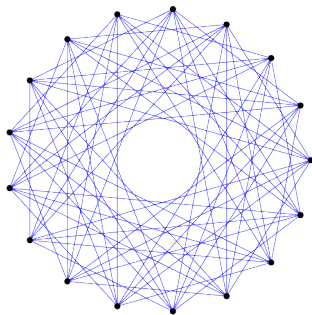
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The blue graph we used to find a lower bound for $R(4, 4) = 18$ is a Paley graph. Treat each vertex like a number modulo 17. Connect two vertices a and b if $a - b$ is a square modulo 17.



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Of the over 2.46×10^{26} edge-colorings of K_{17} , the Paley graph above is the *only* one that works as a lower bound for $R(4, 4)$.

Why are Ramsey numbers difficult to compute?

Theorem (Erdős)

For all natural numbers $t \geq 2$, $R(t, t) > \frac{t \cdot 2^{\frac{t-1}{2}}}{e}$.

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Proof.

Color each edge of K_n red with probability $1/2$.

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Color each edge of K_n red with probability $1/2$. The expected # of monochromatic copies of K_t is $2^{1-\binom{t}{2}} \binom{n}{t}$.

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Color each edge of K_n red with probability $1/2$. The expected # of monochromatic copies of K_t is $2^{1-\binom{t}{2}} \binom{n}{t}$. Take $n = \frac{t \cdot 2^{\frac{t-1}{2}}}{e}$.

$$2^{1-\binom{t}{2}} \binom{n}{t} < 2^{1-\binom{t}{2}} \frac{n^t}{t!} \leq 2^{1-\binom{t}{2}} \left(\frac{t 2^{(t-1)/2}}{e} \right)^t \frac{1}{2} \left(\frac{e}{t} \right)^t = 1$$

The expectation is smaller than 1, so there must be a coloring of K_n that does not contain a monochromatic K_t . □

Known Ramsey numbers

$r \backslash s$	1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	1	1	1	1	1	1
2		2	3	4	5	6	7	8	9	10
3			6	9	14	18	23	28	36	40–41 ^[14]
4				18	25 ^[9]	36–40	49–58	59 ^[15] –79	73–105	92–135
5					43–46 ^[11]	59 ^[16] –85	80–133	101–193	133–282	149 ^[15] –381
6						102–160	115 ^[15] –270	134 ^[15] –423	183–651	204–944
7							205–492	219–832	252–1368	292–2119
8								282–1518	329–2662	343–4402
9									565–4956	581–8675
10										798–16064

Source: Wikipedia, “Ramsey’s theorem”

Recent progress

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Lower bounds for $R(6, 6)$ and $R(8, 8)$ have not been improved since 1965 and 1972, respectively.

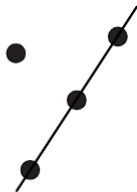
General position

Definition

A set of points in the plane is in *general position* if no three points lie on the same line.



*points in
general position*



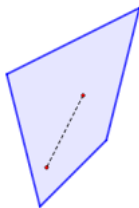
*points **not** in
general position*

Convex polygon

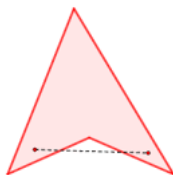
Definition

A polygon is *convex* if the line segment between any two points in the polygon is contained in the polygon.

CONVEX



CONCAVE

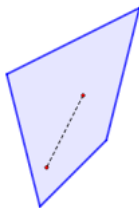


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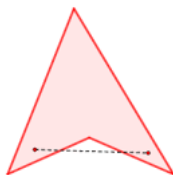
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A polygon is *convex* if the line segment between any two points in the polygon is contained in the polygon.

CONVEX



CONCAVE



Example

If we place three points in general position in the plane, then the points are the vertices of a convex triangle.

Convex quadrilateral

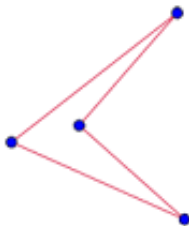
Example

If we place four points in general position in the plane, are the points necessarily the vertices of a convex quadrilateral?

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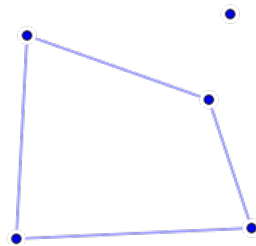
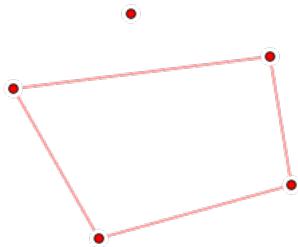
Example

If we place five points in general position in the plane, do four of the points necessarily form the vertices of a convex quadrilateral?

Convex quadrilateral

Example

If we place five points in general position in the plane, do four of the points necessarily form the vertices of a convex quadrilateral?



Happy ending problem

Theorem (Klein)

Any set of five points in the plane in general position has a subset of four points that form the vertices of a convex quadrilateral.

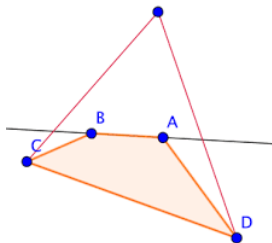
Happy ending problem

Proof.

If the five points form the vertices of a convex pentagon, we are done. Likewise, if four of the points form the vertices of a convex quadrilateral with the fifth point in the middle, we are done.

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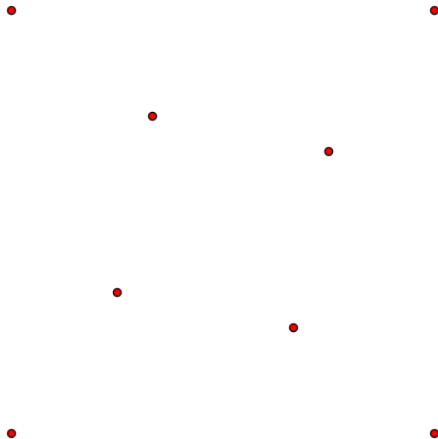
Happy ending problem

Example

Is there a smallest M for which any set of M points in the plane in general position has n points that form vertices of a convex n -gon?

n -gon	M
3	3
4	5
5	
6	

Happy ending problem



Happy ending problem

The previous image proves $M > 8$.

We can also argue that any 9 points in the plane in general position must contain the vertices of a convex pentagon.

n -gon	M
3	3
4	5
5	9
6	

Happy ending problem



Happy ending problem

n -gon	M
3	3
4	5
5	9
6	17

Happy ending problem

n -gon	M
3	3
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Conjecture (Erdős)

The smallest number of points in the plane in general position that guarantees n points form the vertices of a convex n -gon is $2^{n-2} + 1$.

Happy ending problem

n -gon	M
3	3
4	5
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Conjecture (Erdős)

The smallest number of points in the plane in general position that guarantees n points form the vertices of a convex n -gon is $2^{n-2} + 1$.

Proposition (Erdős–Szekeres)

There is a way to arrange 2^{n-2} points in the plane so that no n of the points form the vertices of a convex n -gon.

Happy ending problem

Theorem (Happy ending problem)

For any natural number n , there exists M such that any set of M points in general position in the plane contains n points that form the vertices of a convex n -gon.

Their original proof relies on hypergraph Ramsey numbers.

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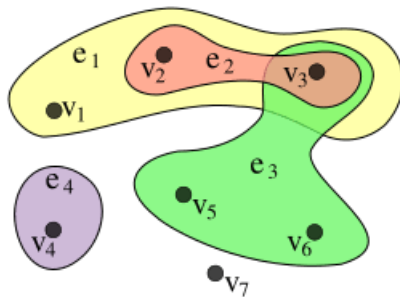
Their original proof relies on hypergraph Ramsey numbers.

The result is known as the “Happy ending problem” because it led to the marriage of Klein and Szekeres.

Definition of a hypergraph

Definition

A *hypergraph* $G = (V, E)$ is a collection of *vertices* and *hyperedges* or subsets of vertices of any size.



Complete k -uniform hypergraphs

Definition

A hypergraph is k -uniform if each hyperedge has size k .

Example

A graph is a 2-uniform hypergraph.



Complete k -uniform hypergraphs

Definition

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Example

A graph is a 2-uniform hypergraph.



Definition

A complete k -uniform hypergraph on n vertices, denoted $K_n^{(k)}$, has every subset of k vertices as a hyperedge.

Hypergraph Ramsey numbers

Theorem (Hypergraph Ramsey's Theorem)

For any natural numbers s , t and k at least two, there exists an N such that every red/blue-coloring of the edge set of $K_N^{(k)}$ contains either a blue copy of $K_s^{(k)}$ or a red copy of $K_t^{(k)}$.

Hypergraph Ramsey numbers

Theorem (Hypergraph Ramsey's Theorem)

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Definition

The smallest N for which Ramsey's Theorem holds is the *hypergraph Ramsey number* $R_k(s, t)$.

Proof of the happy ending problem

Theorem (Happy ending problem)

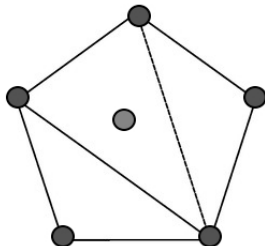
For any natural number n , there exists M such that any set of M points in general position in the plane contains n points that form the vertices of a convex n -gon.

Lemma

If every subset of 4 points forms a convex quadrilateral, then n points in general position form a convex n -gon.

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Assume that one of the n points x lies in the interior of a convex m -gon. Triangulate the m -gon. Since the points are in general position, x is contained in a convex triangle.



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Theorem (Happy ending problem)

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Proof of the happy ending problem.

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Schur's Theorem

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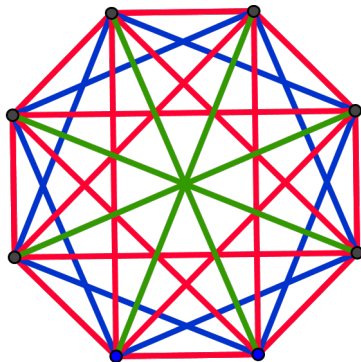
Example

Color a natural number blue if it is congruent to 0 modulo 3, red if it is congruent to 1 modulo 3, or green if it is congruent to 2 modulo 3. Then $3 + 6 = 9$ within the blue color class. However, there is no triple that works in either of the other two colors.

Multicolored Ramsey number

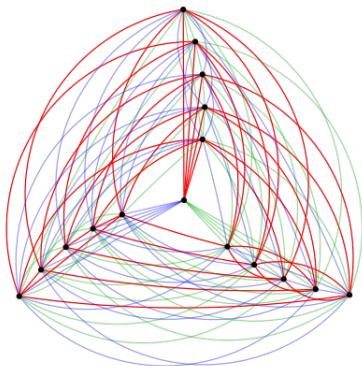
Definition

The *multicolor Ramsey number* $R(s_1, \dots, s_\ell)$ is the smallest N for which K_N must contain some K_{s_i} of color i when the edges of K_N are ℓ -colored.



Example

We can argue that $R(3, 3, 3) = 17$. If we color each edge of K_{17} blue, red, or green, there will be a blue, red, or green triangle. The following is a 3-coloring of the edges of K_{16} that does not contain a monochromatic triangle.



Multicolor Ramsey's Theorem

Theorem (Multicolor Ramsey's Theorem)

There exists a least positive integer $R(s_1, \dots, s_\ell)$ for which every ℓ -coloring of the edges of the complete graph on $R(s_1, \dots, s_\ell)$ vertices contains some K_{s_i} of color i .

Proof of Schur's Theorem

Proof of Schur's Theorem.

Let c denote an ℓ -coloring of the natural numbers. Construct the complete graph K_n on the first $n = R(3, \dots, 3)$ natural numbers. Define an ℓ -coloring c' of the edges of K_n by

$$c'(ij) = c(|j - i|).$$

There exists a monochromatic triangle made up of $u < v < w$ with

$$\begin{aligned} c(v - u) &= c(w - v) = c(w - u) \\ (v - u) + (w - v) &= (w - u) \end{aligned}$$



Questions?

Thank you for your time!

If you have questions or comments about anything, do not hesitate to contact me at mgherman@caltech.edu.